

23 June P3 & Calculus recall sheet

How to use this sheet

Recall the method before opening the answer. Labels distinguish what was reached in the lesson

(paper-anchored or lesson evidence) from official-reference completion and genuinely unfinished work.

Method prompts

P3 Q1(a) — range of f [official WMA13/01 Oct 2025; 1 mark]

- $f(x)=\ln(x^2+3)$, $x \in \mathbb{R}$. The domain is given; the question asks for the range.
- What is the smallest value of x^2+3 ? Take $\ln$ (which is increasing).
- Answer: $f(x) \geq \ln 3$ (range $[\ln 3, \infty)$). The original $f(x)$ in $\mathbb{R}$ was a domain slip, corrected live.

P3 Q1(b) — inverse g^{-1} [official WMA13/01 Oct 2025; 3 marks]

- $g(x)=(5x+3)/(x+2)$, $x > -2$. Rearrange to $x = \dots$, then swap to $g^{-1}(x)$.
- For the domain of g^{-1} , find the range of g : write $g(x)=5-7/(x+2)$ with $x > -2$.
- Answer: $g^{-1}(x)=(3-2x)/(x-5)$, $x < 5$. The missing domain was repaired later at 00:39:43 during F2.

P3 Q1(c) — $fg(0)$ [official WMA13/01 Oct 2025; 2 marks]

- Evaluate $g(0)$, then apply f . Inserted out of order (before finishing (d)).
- Answer: $g(0)=3/2$, so $fg(0)=\ln((3/2)^2+3)=\ln(21/4)$.

P3 Q1(d) — exact a [DEFERRED / UNFINISHED — official reference only]

- $g(e^{\{2a\}})=f(\sqrt{e^4-3})$. The lesson abandoned this after a composition error; not solved live.
- Official reference answer (NOT taught): $a=1/2 \ln 5$. Do not treat this as work you completed.

W0 — warm-up quadratic meets line [lesson evidence; paper unresolved]

- $y=x^2+ax+b$ meets $y=x+1$ at $x=2$ and $x=4$. Equate and substitute the two x -values.
- Answer: $a=-5$, $b=9$. No official mark total (warm-up, source paper unidentified).

NM1 — root location & iteration [lesson evidence; paper unresolved]

- Show a sign change between the ends of $[3,4]$, and state the function is continuous there.
- Iterate $x_{n+1}=F(x_n)$ from $x_1=3$ until the value repeats.
- Answer: root α in $[3,4]$, $\alpha \approx 3.1589$ (4 d.p.). Exact printed function / marks unresolved.

F2 — inverse graph & fixed points [lesson evidence; paper unresolved]

- The recovered function is $f(x)=12/(x+1)$, $x>0$. Reflection in $y=x$ means solve $f(x)=x$.
- Reject $x=-4$ (domain $x>0$); state $x=3$. Inverse domain = range of f .
- Answer: fixed point $x=3$; inverse domain $0<x<12$. Not an official WMA13/01 question.

T1 — prove $\sin 3x$ identity [lesson evidence; paper unresolved]

- $\sin 3x = \sin(2x+x)$, with $\sin 2x=2 \sin x \cos x$ and $\cos 2x=1-2 \sin^2 x$.
- Answer: $\sin 3x = 3 \sin x - 4 \sin^3 x$. Recall the double-angle forms — they are NOT in the booklet.

T2 — 'hence' integral [DEFERRED / UNFINISHED — lesson evidence]

- Reuse T1: $\sin^3 x = 1/4 (3 \sin x - \sin 3x)$, so $\int \sin^3 x \, dx = 1/4 \int (\sin 3x + 3 \sin x) \, dx$.
- The lesson ended before any integral was produced; the antiderivative bracket was left empty.
- No completed integral is supplied.

Status key

- Official WMA13/01 Oct 2025: only Q1(a)-(d) — wording, parts, marks and answers are governed by the identified paper.
- Lesson evidence — paper unresolved: W0, NM1, F2, T1, T2 — genuine lesson items; exact paper/marks not identified, nothing guessed.
- DEFERRED / UNFINISHED: Q1(d) and T2 — started but not completed live; reference answers are shown separately, never as taught completion.
- Reference completion: official result retained for checking, not completed aloud.